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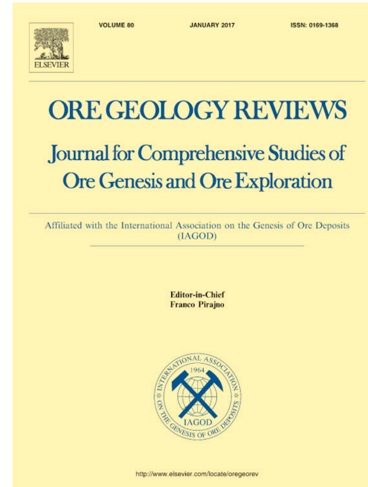

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Gallium- and germanium-rich sphalerite in the Sichuan-Yunnan-Guizhou region: Insights from the Fule carbonate-hosted Pb-Zn deposit

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Abstract

The Sichuan-Yunnan-Guizhou (SYG) region hosts significant resources of critical metals gallium (Ga) and germanium (Ge), which are essential to the development of green technologies. The content and distribution of Ga and Ge in sphalerite from the carbonate-hosted Pb-Zn Fule deposit in this region need to be evaluated. Based on laser ablation-inductively coupled plasma-mass spectrometry analysis, we investigate Ga, Ge, and other minor and trace element composition in different types of sphalerite from this deposit. Two major mineralization stages (I and II) were recognized in hydrothermal period. The highest Ga (mean 218 ppm) and Ge (mean 760 ppm) contents were both found in reddish-brown sphalerite (Sp1*rb*) formed in stage I. Varicolored sphalerite (Sp2) deposited in stage II and white sphalerite (Sp1*w*) formed in stage I exhibit the lowest Ga (mean 0.25 ppm) and Ge (mean 13.2 ppm) contents, respectively. Trace element content correlations reveal Ga incorporation via direct substitution and Ge incorporation via substitution coupled to monovalent copper ion. Additionally, the sphalerite color variations (reddish-brown, yellow, and white) at Fule are related to the contents of Cu, Ga, Ge, As, and Sb. Compared to sphalerite in the other typical Pb-Zn deposits in the SYG region, sphalerite from the Fule deposit is more enriched in Ga and Ge. Considering the close spatial relationship between the Fule deposit and the Emeishan basalts, we propose that the Emeishan basalts may have contributed additional Ga and Ge to this deposit.

Key words: Gallium and germanium; Carbonate-hosted Pb-Zn deposits; Sphalerite color; Emeishan basalts; Sichuan-Yunnan-Guizhou region

1. Introduction

Gallium (Ga) and Germanium (Ge) are considered critical metals due to their high economic importance and supply scarcity (Gulley et al., 2018; U.S. Geological Survey, 2023; European Commission, 2023; Wang et al., 2024). Gallium-based compounds formed with arsenic and nitrogen (i.e., GaAs and GaN) are premium semiconductor materials, which has resulted in the consumption of Ga in the semiconductor industry accounting for an impressive 80% of the total global usage (Moskalyk, 2003; Frenzel et al., 2017). Germanium is mainly employed in infrared optical devices, photovoltaic cells, fiber-optic systems, and as a catalyst in plastic production (Cook et al., 2015). The average contents of Ga and Ge in the continental crust are 15 ppm and 1.5 ppm, respectively (Hans Wedepohl, 1995). Gallium and Ge, which rarely form independent deposits, can be recovered as by-products of zinc mining in carbonate-hosted deposits, volcanic-hosted massive sulfide deposits, sedimentary-exhalative deposits or Kipushi-type deposits (Höll et al., 2007; Frenzel et al., 2014; Licht et al., 2015; Mondillo et al., 2018; Torró et al., 2023).

Gallium and Ge are generally enriched in the sphalerite lattice, typically occurring in the trivalent oxidation state (Ga^{3+}) and tetravalent oxidation state (Ge^{4+}) respectively (Cook et al., 2015; Belissont et al., 2016; Liu et al., 2023). Their incorporation may occur via substitution coupled to monovalent cations (e.g., Cu and Ag; Belissont et al., 2014; Bonnet et al., 2016; Wei et al., 2018; Bauer et al., 2019; Cugerone et al., 2021) and divalent cations (e.g., Pb, Mn, Fe, and Cd; Hu et al., 2021; Luo et al., 2022; Bai et al., 2024). Hydrothermal experiments and simulation calculation show that Ge^{4+} can readily enter the sphalerite lattice, either with or without the need for charge balancing by other metal ions (Liu et al., 2023). Additionally, Ga and Ge may also occur as micrometer- or nanometer-sized independent mineral inclusions in sphalerite (e.g., Cugerone et al., 2020, 2021; Fougerouse et al., 2023; Niu et al., 2024; Sun et al., 2024).

The concentrations of Ga and Ge exhibit significant differences in sphalerite from various types of Zn-Pb sulfide deposits, which are related to fluid temperature (Frenzel et al., 2016). Even at a deposit scale, the enrichment of Ga and Ge in sphalerite demonstrates great heterogeneity. Cook et al. (2015) observed that Ge is richer in darker Fe-rich sphalerite from the Tres Marias deposit, Mexico. Bonnet et al. (2016) revealed that Ga and Ge are respectively enriched in yellow and brown sphalerite from the East Tennessee Mississippi Valley-type deposits, USA. In the Fankou deposit (China), Ga prefers to be enriched in the late-staged and light-colored, sphalerite (Wu et al., 2024). To further the understanding of economic potential and exploration strategies for Ga and Ge, it is essential to ascertain their spatial distribution in sphalerite.

The Sichuan-Yunnan-Guizhou (SYG) region hosts over 400 carbonate-hosted Pb-Zn deposits (Zhang et al., 2015) and these deposits contain significant critical metal resources such as Ga, Ge, and In (Hu et al., 2020; Liu et al., 2022; Zheng et al., 2023). As the only large-scale igneous event between the Ediacaran and Triassic in the SYG region, eruption of the late Permian Emeishan basalts was considered to be related to the above Pb-Zn mineralization (Xu et al., 2014). The Fule deposit in the SYG region, comprising approximately 10 Mt of sulfide ores at mean grades of 15–20 wt.% Zn + Pb, is spatially associated with the Emeishan basalts (Zhou et al., 2018). Bulk trace element analysis has demonstrated that Ga and Ge are enriched in sphalerite from the Fule deposit (Si et al., 2006, 2011). However, their spatiotemporal distribution and substitution mechanism in the sphalerite remains unclear.

An investigation into the minor and trace element contents of different types of sphalerite in the Fule deposit was conducted using laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) analysis. This study deciphers the distribution characteristics and substitution mechanism of Ga and Ge in sphalerite from the Fule deposit. Additionally, the poorly understood possible correlation between trace element contents and sphalerite crystal color is explored. Finally, we present a comparison with other Pb-Zn deposits in the SYG region and endeavour to establish the connection between the Emeishan basalts and the enrichment of critical metals Ga and Ge.

2. Region geology

The SYG region, located on the southwestern margin of the Yangtze Block (Fig. 1a), hosts a remarkable concentration of base metals (> 20 Mt; Zhang et al., 2013) in a prospective area of approximately 170,000 km^2. This region is made of the metamorphic basement and sedimentary cover, displaying an angular unconformity between the two (Liu and Lin, 1999). The metamorphic basement is composed of Mesoproterozoic to Neoproterozoic sedimentary successions including the Dongchuan and Kunyang, Huili Groups, which underwent metamorphism during the Jinning Orogeny (Zhang et al., 2006; Wang and Zhou, 2014). The sedimentary cover mainly consists of the Ediacaran to Triassic marine carbonate rocks, interbedded with clastic rocks and argillaceous rocks (Yan et al., 2003; Yu et al., 2020). Magmatism in SYG region is primarily represented by the late Permian Emeishan basalts, which are widely distributed and up to ~5 km maximum thickness (Fig. 1b; Xu et al., 2014). Basic intrusive rocks sporadically exposed in the southeast of the region (Fig. 1b), and U-Pb dating of zircons reveals approximately 268 Ma for them (Qi et al., 2016).

Structurally, the framework of the SYG region is dominated by the SN-trending Anninghe and Xiaojiang faults, the NW-trending Mile-Shizong fault, and the NE-trending Ziyun-Yadu fault, with secondary faults trending NE, NW, and nearly SN (Fig. 1b). The structural combination formed by the secondary faults and related folds controls the distribution of Pb-Zn mineralization in the region (Han et al., 2014, 2022). As a world-class base metal metallogenic belt, the SYG region contains more than 400 Pb-Zn deposits, which are mostly hosted within Ediacaran to Permian carbonate rocks (Zhang et al., 2015). Among these, notable deposits with Pb + Zn metal resources exceeding 0.5 Mt include Huize, Maoping, Daliangzi, Fule, Tianbaoshan, Maozu, Jingshachang, Maliping, Lehong, Zhugongtang, and Chipu (Fig. 1b). Furthermore, the Pb-Zn deposits in SYG region contain not only significant amounts of Pb and Zn but are also enriched with critical metals such as Ga, Ge, and In (Meng et al., 2024 and references therein).

The Ga and Ge background contents of the metamorphic basement rocks, the sedimentary cover rocks, and the Emeishan basalts in the SYG region are shown in Fig. 2. The Ga content in the Emeishan basalts is higher than that in the metamorphic basement rocks and the sedimentary cover rocks, as well as the Ga average content in continental crust (Fig. 2a). On the other hand, the Ge content in the basement rocks is higher than that in the sedimentary cover rocks and the Emeishan basalts, but all of their Ge contents are lower than the Ge average content in continental crust (Fig. 2b).

3. Ore deposit geology

The strata exposed in the Fule mining area consist of Permian to Triassic sedimentary succession (Fig. 3a). The middle Permian Yangxin Formation is represented by an interbedded gray to dark gray limestone and dolostone, which is unconformably overlain by late Permian Emeishan basalts. The upper Permian Xuanwei Formation is mainly composed of grayish-green calcareous shale and black carbonaceous shale, intercalated with coal seams. The lower Triassic Feixianguan Formation consists of sandstone, siltstone, and shale, and is overlain by the Yongning Formation which is dominated by gray argillaceous limestone. The middle Triassic Guanling Formation is the uppermost succession of the area and comprises sandstone, dolomite, and limestone. Structurally, the area is dominated by NNE- and nearly SN-trending reverse faults (Fig. 3a).

More than 20 orebodies are hosted in the Middie Permian Yangxin Formation, and small orebodies are distributed in a "satellite" pattern around the large ones (Ren et al., 2019). Orebodies occur as lenses, veins, and stratiform shapes (Fig. 3b). The sulfide ores are predominantly composed of sphalerite, galena, and pyrite, with Cu and Ni sulfide minerals such as chalcopyrite, tetrahedrite, millerite, niccolite, pentlandite, polydymite (Li et al., 2018a, b; Zhou et al., 2018; Chen et al., 2022). Common secondary metallic minerals in the oxidized ores include smithsonite, hemimorphite, cerussite, and malachite (Si, 2005). Calcite and dolomite occur as main gangue minerals. The ore structures primarily include disseminated, brecciated, massive, and stockwork filling structures and the ore textures mainly display granular, metasomatic, solid solution, and cataclastic textures (Zhou et al., 2018; Cui et al., 2018).

The ore-forming process of the Fule deposit can be divided into diagenetic, hydrothermal and supergene periods (Fig. 4). As suggested by Si (2005) and Zhou et al. (2018), two mineralization stages were recognized in the hydrothermal period: sulfide-carbonate stage (stage I) and carbonate stage (stage II). Stage I includes sphalerite-calcite/dolomite and sphalerite-galena-calcite/dolomite mineral assemblages. Stage II is dominated by abundant calcite and dolomite, associated with minor sphalerite and galena.

4. Materials and methods

4.1. Sample

Six samples representing the hydrothermal period were collected from underground galleries and drill cores at Fule. Polished sections (150 μm thick) for all samples were prepared for examination using optical microscope. Sphalerite formed in the stage I (Sp1) occurs mostly as coarse-grained crystals, associated with calcite (Fig. 5a). Based on color under transmitted light, subhedral Sp1 can be classified into the reddish-brown (Sp1*rb*), yellow (Sp1*y*), and white (Sp1*w*) types (Fig. 5c). Sphalerite related to the stage II (Sp2) is fine-grained and appears as disseminated clusters within calcite (Fig. 5b).

Under transmitted light, euhedral to subhedral Sp2 mostly appears variegated, generally with a darker color at its center and a lighter color towards the edges (Fig. 5d).

4.2. Analytical methods

Spot analyses of trace element in sphalerite were conducted using a 193 nm ArF excimer laser-ablation system coupled to an Thermo Scientific iCap-RQ (ICP-MS) instrument at Guangzhou Tuoyan Analytical Technology Co., Ltd., Guangzhou, China. Helium was used as the carrier gas with Ar mixed in as the makeup gas to adjust sensitivity during the laser ablation process. A laser energy density was 4 J/cm^2, with a repetition rate of 6 Hz, and a spot diameter of 35 μm. The following isotopes were monitored during each measurement (40 s background, 45 s ablation): ^{55}Mn, ^{57}Fe, ^{65}Cu, ^{71}Ga, ^{74}Ge, ^{75}As, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{202}Hg, ^{204}Tl and ^{208}Pb. Trace element composition of sphalerite was calibrated against the reference materials NIST 610, MASS-1, and BCR-2G without using an internal standard (Liu et al., 2008). The reference material NIST 612 was used as a monitoring standard. Raw data processing was completed using a software Iolite (Paton et al., 2011).

Trace element mapping of sphalerite was performed using the same LA-ICP-MS system as for the spot analyses. During the mapping, a laser energy of 5 J/cm^2 and a laser repetition rate of 15 Hz were set, with a spot diameter of 10 μm and a scan speed of 40 μm/s. The reference material NIST 610 was used as an external standard.

5. Results

5.1. LA-ICP-MS spot analyses

A total of 80 spot analyses were conducted and trace element composition of sphalerite is summarized in Table 1 and Fig. 6. In the results, Values below detection limit (b.d.l.) were replaced with respective detection limits for evaluation of minimum, maximum, mean, median, and interquartile range values (e.g., Frenzel et al., 2022). When an element is missing more than 50% of its all values, it is omitted from the data analysis and interpretation (cf. Frenzel et al., 2023). The complete LA-ICP-MS trace element dataset including detection limits is presented in Electronic Appendix A. No peak signals were observed in LA-ICP-MS time resolution profiles (Electronic Appendix B; Fig. S1), indicating the selected elements may respond to solid solutions within the sphalerite, rather than mixed micro-mineral inclusions (e.g., Cook et al., 2009; Ye et al., 2011).

Cadmium is significantly rich in sphalerite from the Fule deposit and is the only element that reaches contents higher than 1 wt.%. The highest Cd contents of 16111 to 40201 ppm occur in Sp1*w* (mean 26439 ppm, median 21929 ppm). The lowest Cd contents were recorded in Sp1*rb*, varying from 13417 to 21476 ppm (mean 16933 ppm, median 16346 ppm). Note that an abnormal value of Cd content up to 48492 ppm occurs in Sp2.

The darker Sp1*rb* hosts the lowest Fe contents (2115 to 2669 ppm, mean 2351 ppm, median 2330 ppm), while the lighter Sp1*y* contains the highest Fe contents (2175 to 3531 ppm, mean 3082 ppm, median 3221 ppm). Copper is the other element that has contents higher than 1000 ppm in sphalerite from the Fule deposit. The highest Cu contents were found in Sp1*rb*, varying from 1919 to 3143 ppm (mean 2709 ppm, median 2753 ppm).

As for critical elements, the sphalerite is commonly enriched in Ga and Ge. The highest Ga contents occur in Sp1*rb*, varying from 58.2 to 474 ppm (mean 218 ppm, median 214 ppm). Gallium contents in Sp1*rb*, Sp1*y* (2.76 to 386 ppm, mean 101 ppm, median 83.5 ppm), and Sp1*w* (b.d.l. to 251 ppm, mean 41.5 ppm, median 19.0) exhibit a decreasing trend. Nonetheless, Sp2 is deficient in Ga, with a range of b.d.l. to 1.26 ppm (mean 0.25 ppm, median 0.08 ppm).

The highest Ge contents were obtained for Sp1*rb*, varying from 535 to 1038 ppm (mean 760 ppm, median 716 ppm). The lowest contents were recorded in Sp1*w*, with a range of b.d.l. to 30.6 ppm (mean 13.2 ppm, median 11.7 ppm). Germanium contents vary significantly in Sp2, ranging from 0.40 to 488 ppm (mean 150 ppm, medium 122 ppm).

The highest contents of Sb and Pb were both found in Sp1*rb*, with varying ranges of 261 to 929 ppm (mean 568 ppm, median 516 ppm) and 53.5 to 239 ppm (mean 96.0 ppm, median 87.0 ppm), respectively. The lowest contents of Sb and Pb were both obtained in Sp1*w*, with varying ranges of b.d.l. to 18.2 ppm (mean 6.14 ppm, median 4.24 ppm) and 0.07 to 13.3 ppm (mean 3.39 ppm, median 2.21 ppm), respectively. The highest and lowest contents of As were observed in Sp2 and Sp1*w*, respectively, with varying ranges of b.d.l. to 490 ppm (mean 145 ppm, median 124 ppm) and b.d.l. to 4.20 ppm (mean 1.71 ppm, median 1.53 ppm).

Manganese, Ag, and Hg contents in the sphalerite are systematically low, mostly at a few ppm. The highest Mn contents were found in Sp1*w*, varying from 2.64 to 6.11 ppm (mean 3.82 ppm, median 3.62 ppm). The highest Ag contents were recorded in Sp2, varying from 0.23 to 27.7 ppm (mean 7.92 ppm, median 4.66 ppm). The highest Hg contents were found in Sp1*rb*, varying from 15.3 to 244 ppm (mean 54.1 ppm, median 24.2 ppm).

The contents of In, Sn, and Tl in the sphalerite are negligible, with most of them below the detection limits.

5.2. LA-ICP-MS mapping

The LA-ICP-MS mapping images clearly reflect the distribution of trace elements in different color zones of Sp1 (Fig. 7). Copper, Ga, Ge, As, Sb, and Pb are concentrated in Sp1*rb*, Cd is concentrated in Sp1*w*, while Fe is relatively evenly distributed in Sp1. In contrast, Mn, Ag, In, Sn, and Tl all show a depletion in each color zone (Electronic Appendix B; Fig. S2). In Sp1, Ge contents are broadly positively correlated with Cu and Sb (Figs. 7c, e, and h).

For Sp2, the LA-ICP-MS mapping images reveal that Cu, Ge, As, Ag, Cd, Sb, and Pb are concentrated at the center of the crystal grains, while Fe is distributed relatively uniformly throughout the entire crystal grains (Fig. 8). In contrast, Mn, Ga, In, Sn, and Tl are depleted in Sp2 (Figs. 8e and j; Electronic Appendix B; Fig. S3). The growth direction of sphalerite appears to correlate with trace elements (e.g., Bonnet et al., 2016), and Ge is preferentially incorporated along the (010) growth direction in Sp2 (Fig. 8f). In Sp2, Ge contents are broadly positively correlated with Cu (Figs. 8d and f).

6. Discussion

6.1. Incorporation mechanisms of Ga and Ge into sphalerite

Previous studies documented in other deposits have proposed a Ga^{3+} + Cu^{+} (or Ag^{+}) ↔ $2Zn^{2+}$ coupled substitution in sphalerite (e.g., Belissont et al., 2014; Wei et al., 2018; Wu et al., 2024). However, there is no positive correlation between Ga and Ag in sphalerite from the Fule deposit (Fig. 9a). A clear positive correlation (R^2 = 0.88) between Ga and Cu in Sp1*w* is discerned, but these data points significantly deviate from the $(Ga/Cu)_{mol}$ = 1/1 line (Fig. 9b). On the other hand, the depletion of In and Sn in sphalerite from the Fule deposit does not support the coupled substitution of $(Ga, In)^{3+}$ + Cu^{+} + Sn^{4+} ↔ $4Zn^{2+}$ as documented by Torró et al. (2023) and Zhu et al. (2024). Furthermore, fairly positive correlations between Ga and trivalent cations (e.g., Sb^{3+}) are also not observed in the sphalerite (Fig. 9c). Collectively, we suggest a direct substitution of Ga into the sphalerite lattice: $2Ga^{3+}$ + vacancy ↔ $3Zn^{2+}$.

As for Ge, it can be incorporated into sphalerite lattice via common coupled substitutions with Cu^{+} or Ag^{+} (Belissont et al., 2014, 2016; Bauer et al., 2019; Cugerone et al., 2021; White et al., 2022). In sphalerite from the Fule deposit, there is no positive correlation with poorly concentrated Ag (Fig. 9d), similar to Cook et al. (2015). In contrast, both the binary correlation diagram (R^2 = 0.96; Fig. 9e) and the mapping images (Figs. 7 and 8) show a strong positive correlation between Ge and Cu in the sphalerite, suggesting coupled substitutions of the two elements, such as Ge^{4+} + $2Cu^{+}$ ↔ $3Zn^{2+}$. In some cases, the coupled incorporation of Ge with divalent cations has been reported, such as Ge^{4+} + $2Fe^{2+}$ + vacancy ↔ $4Zn^{2+}$ (Yuan et al., 2018; Bai et al., 2024) and Ge^{4+} + M^{2+} + vacancy ↔ $3Zn^{2+}$ (M = Mn or Pb) (Hu et al., 2021; Luo et al., 2022). However, there is no discernible positive correlation between Ge and divalent cations (e.g., Mn^{2+}, Fe^{2+}, and Pb^{2+}) in sphalerite from the Fule deposit (Figs. 9f–h). Additionally, trivalent cations (e.g., Ga^{3+}) also show no correlation with Ge (Fig. 9i).

Remarkably, the possibility that Ga and Ge occur as nano-sized mineral inclusions in sphalerite from the Fule deposit has not been excluded. In some cases, Ge-rich nanoparticles in sphalerite have been recently recognized by transmission electron microscope or atom probe tomography analysis (e.g., Fougerouse et al., 2023; Sun et al., 2024).

6.2. Chemical composition vs. color of sphalerite

Iron is the most common impurity element in sphalerite, and for many years, it is widely considered that the color of sphalerite is directly and inherently determined by its Fe content (Cook et al., 2009; Li and Banes, 2019; Knorsch et al., 2020). Generally, as the Fe content in sphalerite increases, its color gradually changes from colorless to yellow, brown, and even black. For example, sphalerite with an Fe content exceeding 10 wt.% typically appears black in color and is known as marmatite. However, this is not the case for Fe-poor sphalerite, where color variations can be attributed to a complex interplay of multiple impurity elements (Kelley et al., 2004; Bonnet et al., 2016). In our study, the Fe-poor Sp1 (2115 to 3531 ppm; Table 1) exhibits three colors: reddish-brown, yellow, and white. In the following, possible correlations between chemical composition and color of sphalerite are explored.

Cadmium is the impurity element with the highest content in sphalerite from the Fule deposit (Table 1). Principal component analysis (PCA) applied to Sp1, accounting for 79.2 % of element content variability, shows that Cd is correlated with light-colored Sp1*w* (Fig. 10). However, a statistical study by Liu et al. (2015) on the Cd content in sphalerite from 74 deposits worldwide indicates that the Cd content in sphalerite is not related to its color. Although white Sp1*w* contains the highest Cd content, reddish-brown Sp1*rb* and yellow Sp1*y* are also notably enriched in Cd, with all three displaying Cd levels within the same order of magnitude (Table 1). This suggests that Cd does not seem to be the factor for color variation in Sp1. On the other hand, the highest Fe contents were not found in reddish-brown Sp1*rb*, but rather in yellow Sp1*y*, followed by white Sp1*w* (Fig. 6). Elemental mapping of Sp1 also reveals an unusual distribution of Fe that is unrelated to the colors of Sp1 (Figs. 7a and b). Therefore, there is not a systematic correlation between the color and Fe contents in Sp1.

Reddish-brown sphalerite is correlated with high Cd, Cu, Mn, Hg, Ge, and Ag contents in the Eastern Cordillera and sub-Andean regions of Peru (Torró et al., 2023). Among the aforementioned elements, however, only Cu and Ge are relatively high in reddish-brown Sp1*rb* (Fig. 6). Conversely, the PCA shows that reddish-brown Sp1*rb* is correlated with higher Cu, Ga, Ge, As, Sb, and Pb (Fig. 10). These element contents all systematically display a decreasing trend from Sp1*rb* to Sp1*y* to Sp1*w*, thus suggesting that some of them may have an impact on color variation of Sp1. Considering the commonly reported color-causing elements from previous studies (e.g., Belissont et al., 2014; Gagnevin et al., 2014; Bonnet et al., 2016; Yang et al., 2022), the color variation in Sp1 may be jointly attributed to Cu, Ga, Ge, As, and Sb, and high contents of them are correlated with reddish-brown in sphalerite. The correlation between these elements and the three colors is also clearly depicted on elemental mapping of Sp1 (Fig. 7).

6.3. Differential enrichment of Ga and Ge in the SYG region

Previous studies reported that sphalerite in the SYG region is the main carrier of Ge, Ga, Cd and In (Meng et al., 2024 and references therein). In the SYG region, sphalerite from the Fule deposit not only has the highest Cd content (mean 1.9 wt.%; Si et al., 2006), but is also more enriched in Ga and Ge compared to sphalerite in some other typical Pb-Zn deposits, such as Huize and Maoping (Fig. 11). Those Pb-Zn deposits share similarities in geology settings, however, only the Fule deposit demonstrates extreme enrichment of Ga to the economic level. Notably, except for sphalerite, pyrite from the Fule deposit is also enriched in Ge (0.8 to 347 ppm, mean 47.0 ppm; Li et al., 2019), which is rare in the SYG region.

A comprehensive comparison of geological characteristics (Table 2) shows that the spatial relationship between the Fule deposit and the Emeishan basalts is closer than that between the Huize and Maoping deposits and the Emeishan basalts. The Fule deposit is hosted within the middle Permian Yangxin Formation, which is the closest Pb-Zn ore-hosting strata to the Emeishan basalts in the SYG region, and the minimum straight-line distance between its orebody and the Emeishan basalts is < 1 m (Zhou et al., 2018). Nevertheless, primary Pb-Zn mineralization in the SYG region occurred at 230–200 Ma (Hu et al., 2017), much younger than the Permian Emeishan magmatic event (259–263 Ma; Zhou et al., 2002; He et al., 2007), thereby indicating that there is not a directly genetic link between them. Some authors suggest that the contribution of the Emeishan basalts to Pb-Zn deposits in the SYG region could be providing Pb-Zn metals (e.g., Miao et al., 2023) and acting as an impermeable and protective layer (e.g., Li et al., 2012; Zhou et al., 2018).

If the assumption is made that the water-rock reaction took place between the Fule ore-forming fluids and the Emeishan basalts, certain elements may be mobilized from the Emeishan basalts into the hydrothermal fluids. Calcite veins, which are associated with the Pb-Zn mineralization, occur in the Emeishan basalts (Electronic Appendix B; Fig. S4). This observation supports the above assumption. Most carbonate-hosted Pb-Zn deposits rich in Cu-Ni are interpreted in two ways: either as indicative of the presence of leachable Cu-Ni-rich mafic or ultramafic rocks in the source regions (e.g., Horrall et al., 1993), or as reflecting the involvement of high-temperature (> 250℃) fluids in the ore-forming processes (e.g., Wilkinson et al., 2005; Torremans et al., 2018). Fluid inclusion studies revealed that the fluid temperatures are < 250℃ in the Fule deposit (Lyu et al., 2020), suggesting that the fluid did not possess the capability to transport high-temperature elements (e.g., Cu and Ni) over long distances. Indeed, the presence of Cu- and Ni-bearing minerals in the Fule deposit (Table 2) has been attributed to the adjacent Emeishan basalts, which are rich in Cu and Ni (Li et al., 2018a, b). Additionally, the Emeishan basalts are also an important Pb source for the Fule deposit, based on Pb isotope studies (Zhou et al., 2018). It is interesting that the Emeishan basalts generally have the high background values of Ga (24.3–30.7 ppm;

Du et al., 2019). These mean that, despite the substantial contribution of the metamorphic basement to the enrichment of Ga and Ge in the SYG region (Meng et al., 2024), the Emeishan basalts may also serve as a potential source of these metals in the Fule deposit.

7. Conclusions

In this study, LA-ICP-MS trace element analyses on sphalerite from the Fule deposit have been performed. We particularly focused on critical metals Ga and Ge. The conclusions are as follows.

There are significant differences in the contents of Ga and Ge in Fe-poor sphalerite of different types from the Fule deposit, with the contents of Ge generally being higher than those of Ga. In Sp1, the Ga contents are highest in Sp1*rb* (58.2–474 ppm; mean 218 ppm), followed by Sp1*y* (2.76–386 ppm; mean 101 ppm), and lowest in Sp1*w* (b.d.l.–251 ppm; mean 41.5 ppm). In Sp2, however, Ga is depleted (b.d.l.–1.26 ppm; mean 0.25 ppm). The variation in Ge contents in Sp1 is similar to that of Ga, with a systematic decrease from Sp1*rb* (535–1038 ppm; mean 760 ppm) to Sp1*y* (43.1–379 ppm; mean 100 ppm) to Sp1*w* (b.d.l.–30.6 ppm; mean 13.2 ppm). However, Sp2, which is depleted in Ga, is enriched in Ge (0.40–488 ppm; mean 150 ppm).

Comparisons of mapping images and correlations between molar contents suggest a direct substitution of Ga into the sphalerite crystal lattice: $2Ga^{3+}$ + vacancy $\leftrightarrow 3Zn^{2+}$. Germanium incorporation in sphalerite crystal lattice is a coupled substitution with monovalent copper ion, such as $Ge^{4+} + 2Cu^{+} \leftrightarrow 3Zn^{2+}$. The PCA shows that Reddish-brown sphalerite is correlated with higher Cu, Ga, Ge, As, and Sb contents in the Fule deposit.

Gallium and Ge contents in sphalerite from the Fule deposit are notable in the SYG region. The potential acquisition of additional Ga and Ge from the Emeishan basalts could account for abundant presence of these elements in the Fule deposit. Future studies should further explore the role played by the Emeishan basalts in the enrichment of critical metals in the SYG region.

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Figure captions

Fig. 1. (a) Simplified map of South China showing the location of the SYG region. (b) Schematic geological map of the SYG region, showing the main faults and the distribution of main Pb-Zn deposits, including Fule (modified after Liu and Lin, 1999; Luo et al., 2019).

Fig. 2. Background contents of Ga (a) and Ge (b) in the SYG region (data from Mo et al., 2013; Du et al., 2019). The average contents of Ga and Ge in continental crust are from Hans Wedepohl (1995).

Fig. 3. (a) Geological sketch map of the Fule mining area showing the location of the cross-section in Fig. 2b (modified after Liu and Lin, 1999). (b) Geological cross-section of the Fule deposit (modified after Liu and Lin, 1999).

Fig. 4. Paragenetic sequence of the Fule deposit (modified after Si, 2005; Zhou et al., 2018).

Fig. 5. Representative petrographic photos from the Fule deposit. (a) Coarse-grained sphalerite (Sp1) intergrown with calcite. Sp1 displays various colors. (b) Fine-grained sphalerite (Sp2) is disseminated in calcite. (c) Subhedral sphalerite (Sp1), showing reddish-brown (Sp1*rb*), yellow (Sp1*y*), and white (Sp1*w*) zones (transmitted light). (d) Euhedral to subhedral sphalerite (Sp2) embedded in calcite. The center of Sp2 is darker in color, with lighter color at its edges (transmitted light). Abbreviations: Cal-calcite; Gn-galena; Lim-limestone.

Fig. 6. Box plot diagrams for LA-ICP-MS trace element composition in four types of

sphalerite from the Fule deposit.

Fig. 7. LA-ICP-MS trace element mapping images of Sp1. (a) Transmitted light photomicrograph. (b–i) LA-ICP-MS mapping for selected elements (Fe, Cu, Ga, Ge, As, Cd, Sb, and Pb).

Fig. 8. LA-ICP-MS trace element mapping images of Sp2. (a) Transmitted light photomicrograph. (b) Backscattered electron image. (c–l) LA-ICP-MS mapping for selected elements (Fe, Cu, Ga, Ge, As, Ag, Cd, In, Sb, and Pb).

Fig. 9. Binary correlation diagrams for trace elements in sphalerite from the Fule deposit. (a) Ag vs. Ga; (b) Cu vs. Ga; (c) Sb vs. Ga; (d) Ag vs. Ge; (e) Cu vs. Ge; (f) Mn vs. Ge; (g) Fe vs. Ge; (h) Pb vs. Ge; (i) Ga vs. Ge.

Fig. 10. Principal component analysis (PCA) of the log-transformed dataset of trace element contents in sphalerite from the Fule deposit. Left frame: eigenvalues and loadings of the principal components. Middle frame: spot analyses plotted on the PC1 vs. PC2 plane (explaining 79.2% of element content variability). Right frame: loading plot showing the elements with similar behavior.

Fig. 11. Boxplot of LA-ICP-MS data for Ga (a) and Ge (b) in sphalerite from typical Pb-Zn deposits in the SYG region. The Huize and Maoping deposits are based on published LA-ICP-MS data from Oyebamiji et al. (2020) and Wang et al. (2023a), respectively. Fule: Sp1*rb* = reddish-brown sphalerite in stage I, Sp1*y* = yellow sphalerite in stage I, Sp1*w* = white sphalerite in stage I, Sp2 = sphalerite in stage II; Huize: Sp1 = sphalerite in stage I, Sp2 = sphalerite in stage II; Maoping: Sp2*dr* = dark red sphalerite in stage II, Sp2*yb* = yellow-brown sphalerite in stage II, Sp2*ly* = light-yellow sphalerite in stage II.

Table captions

Table 1. Summary of LA-ICP-MS trace element composition of sphalerite from the Fule deposit (data in ppm).

Table 2. Comparison of principal geological characteristics of typical Pb-Zn deposits in the SYG region.

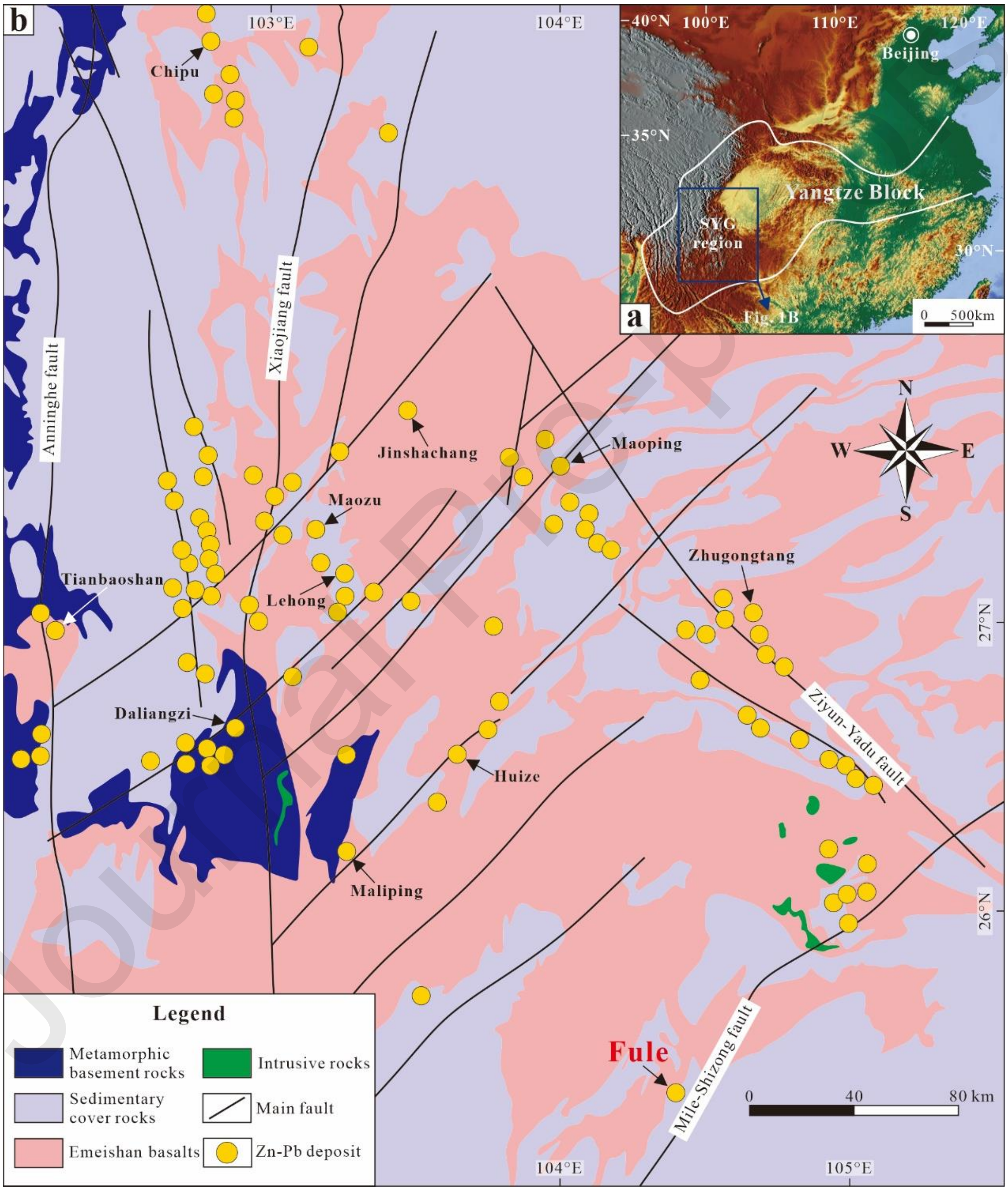
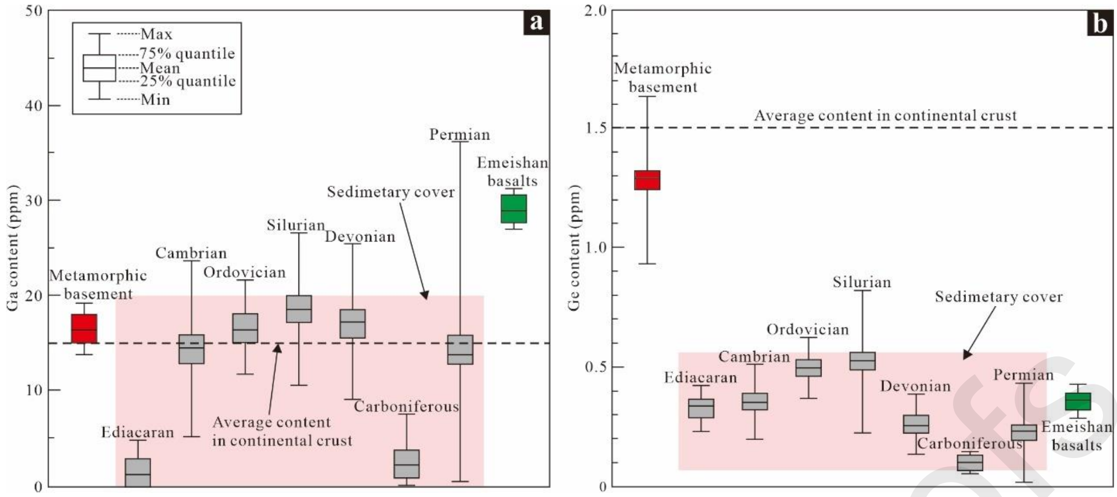
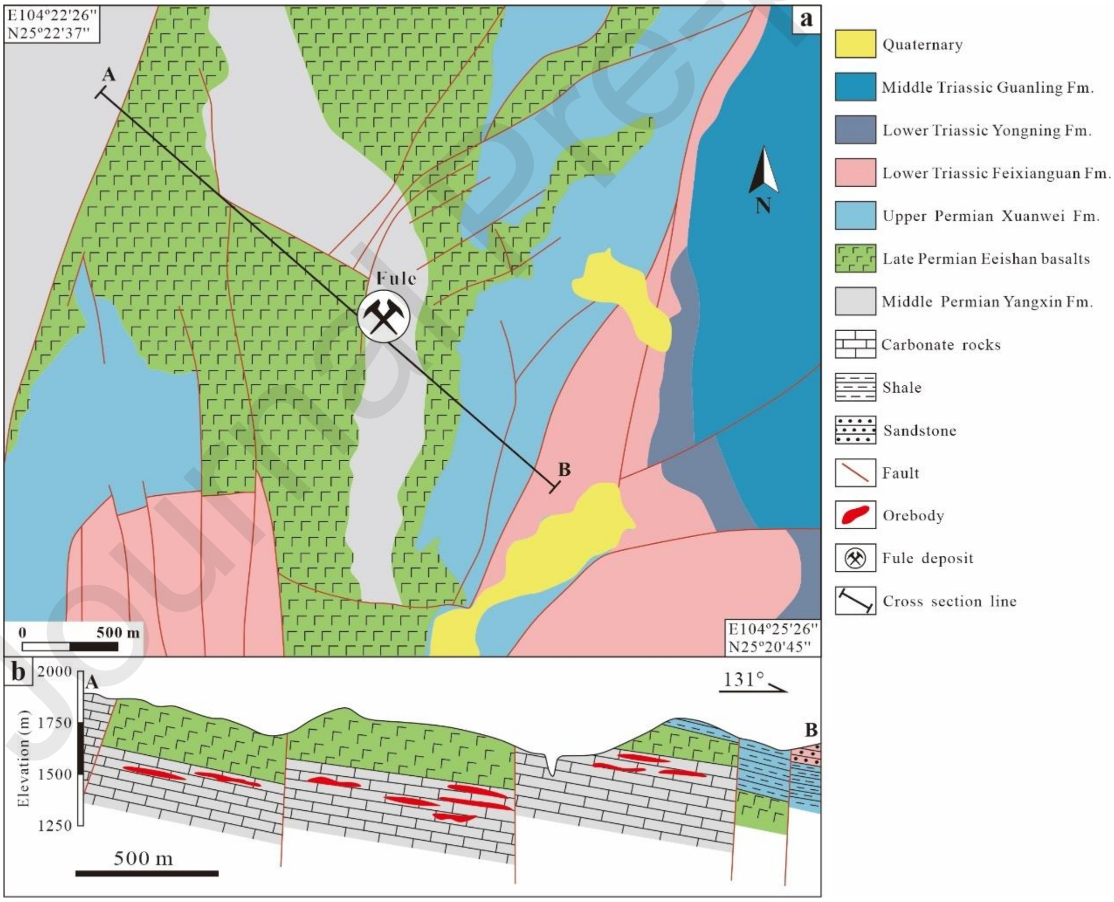


Fig. 1



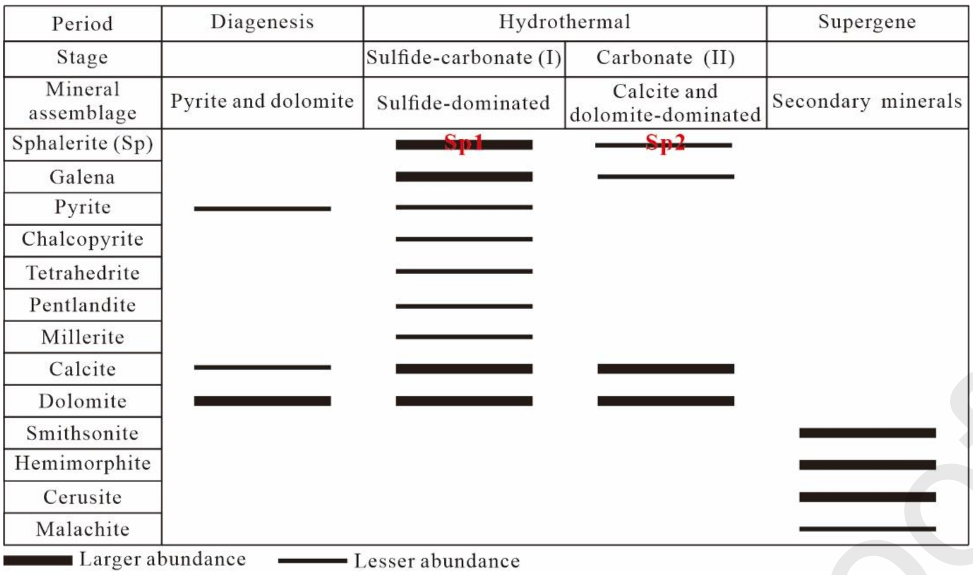
a
Max
75% quantile
Mean
25% quantile
Min
Ga content (ppm)
Metamorphic basement
Ediacaran
Cambrian
Ordovician
Silurian
Devonian
Carboniferous
Permian
Emeishan basalts
Sedimetary cover
Average content in continental crust
b
Ge content (ppm)
Average content in continental crust

Fig. 2



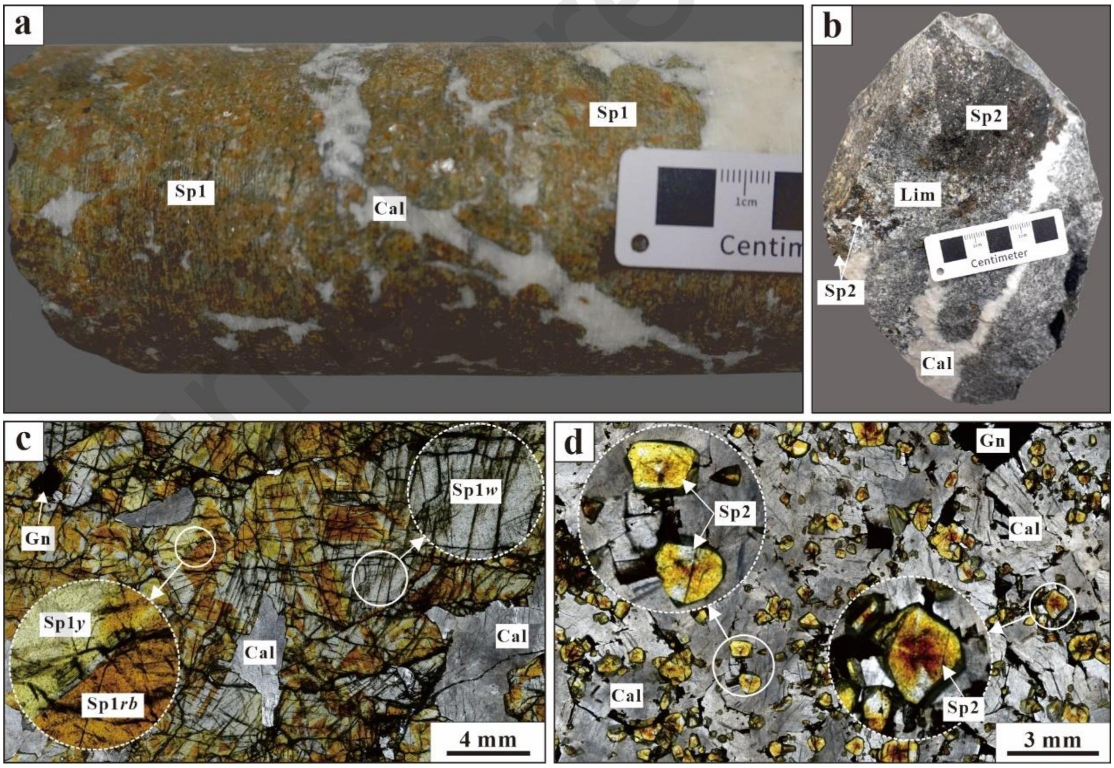
E104°22'26"
N25°22'37"
E104°25'26"
N25°20'45"
a
A
B
Fule
N
0
500 m
Quaternary
Middle Triassic Guanling Fm.
Lower Triassic Yongning Fm.
Lower Triassic Feixianguan Fm.
Upper Permian Xuanwei Fm.
Late Permian Eeishan basalts
Middle Permian Yangxin Fm.
Carbonate rocks
Shale
Sandstone
Fault
Orebody
Fule deposit
Cross section line
b
Elevation (m)
131°
500 m

Fig. 3



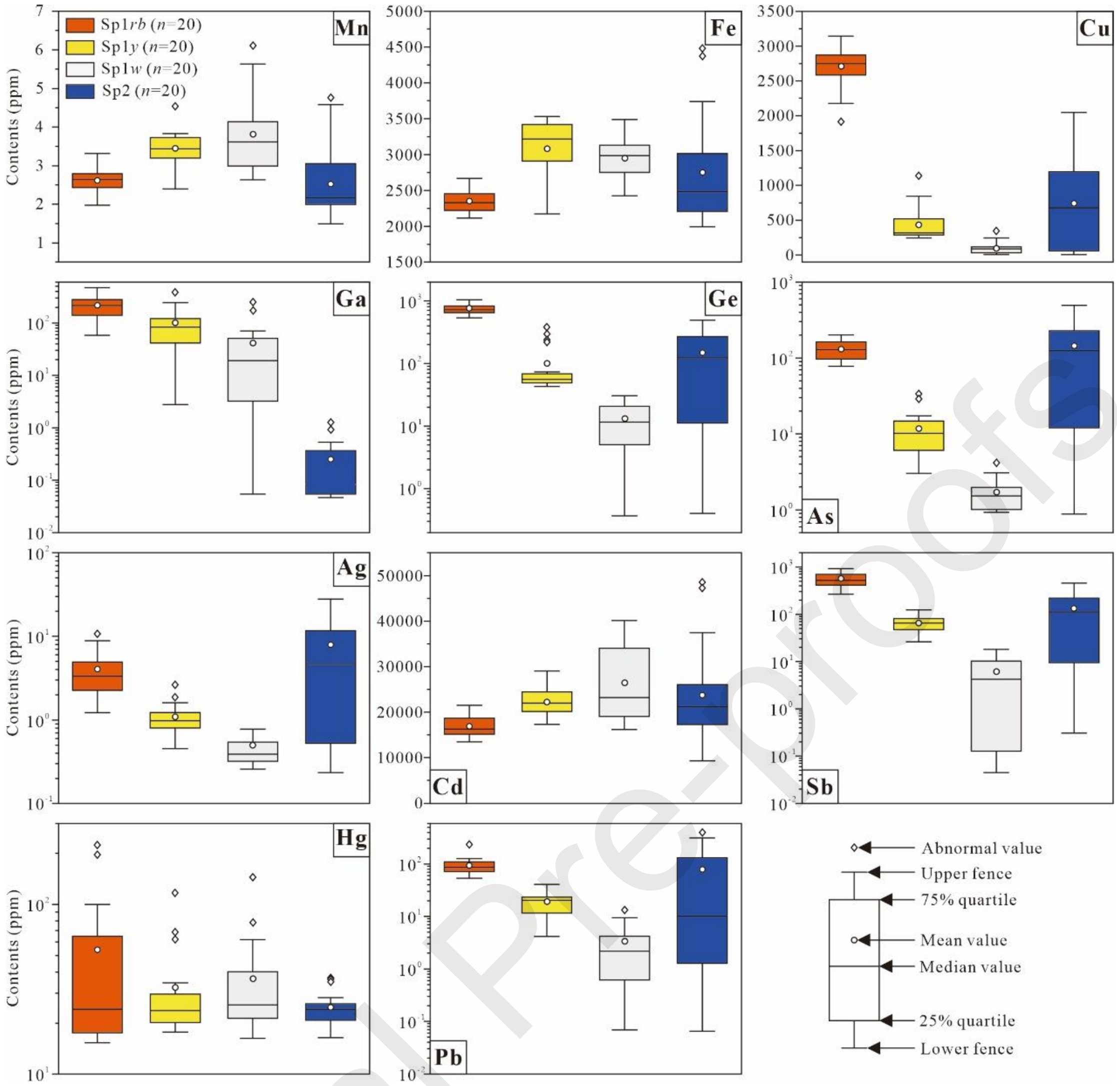
Period
Diagenesis
Hydrothermal
Supergene
Stage
Sulfide-carbonate (I)
Carbonate (II)
Mineral assemblage
Pyrite and dolomite
Sulfide-dominated
Calcite and dolomite-dominated
Secondary minerals
Sphalerite (Sp)
Sp1
Sp2
Galena
Pyrite
Chalcopyrite
Tetrahedrite
Pentlandite
Millerite
Calcite
Dolomite
Smithsonite
Hemimorphite
Cerusite
Malachite
Larger abundance
Lesser abundance

Fig. 4



a
Sp1
Sp1
Cal
b
Sp2
Lim
Sp2
Cal
c
Gn
Sp1w
Sp1y
Sp1rb
Cal
Cal
4 mm
d
Gn
Sp2
Cal
Cal
Sp2
3 mm

Fig. 5



Sp1rb (n=20)
Sp1y (n=20)
Sp1w (n=20)
Sp2 (n=20)
Mn
Fe
Cu
Ga
Ge
As
Ag
Cd
Sb
Hg
Pb
Contents (ppm)
Abnormal value
Upper fence
75% quartile
Mean value
Median value
25% quartile
Lower fence

Fig. 6

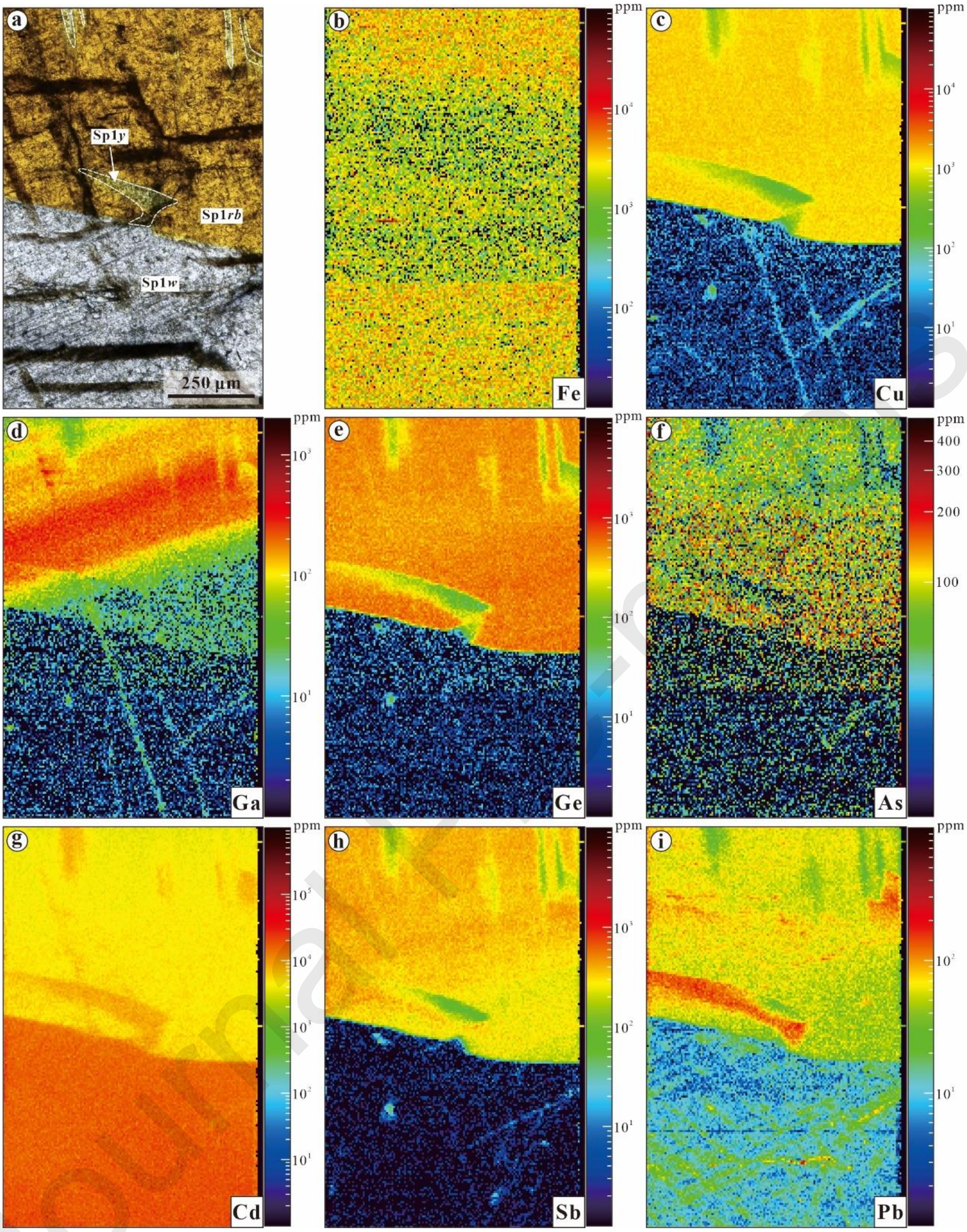
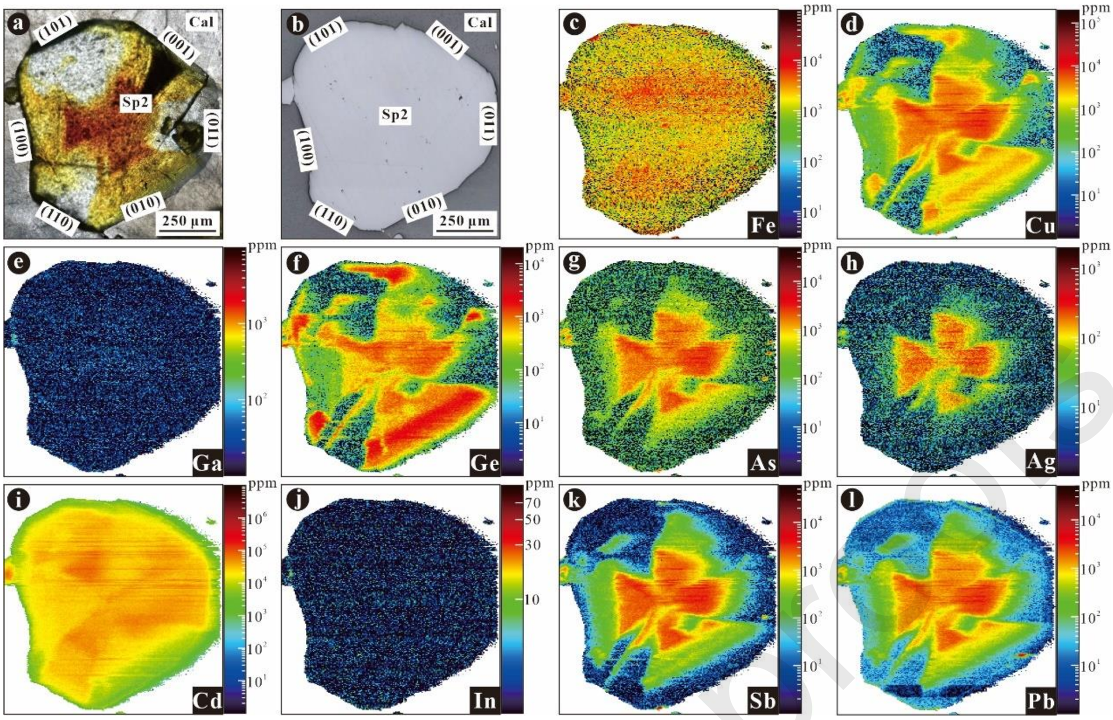
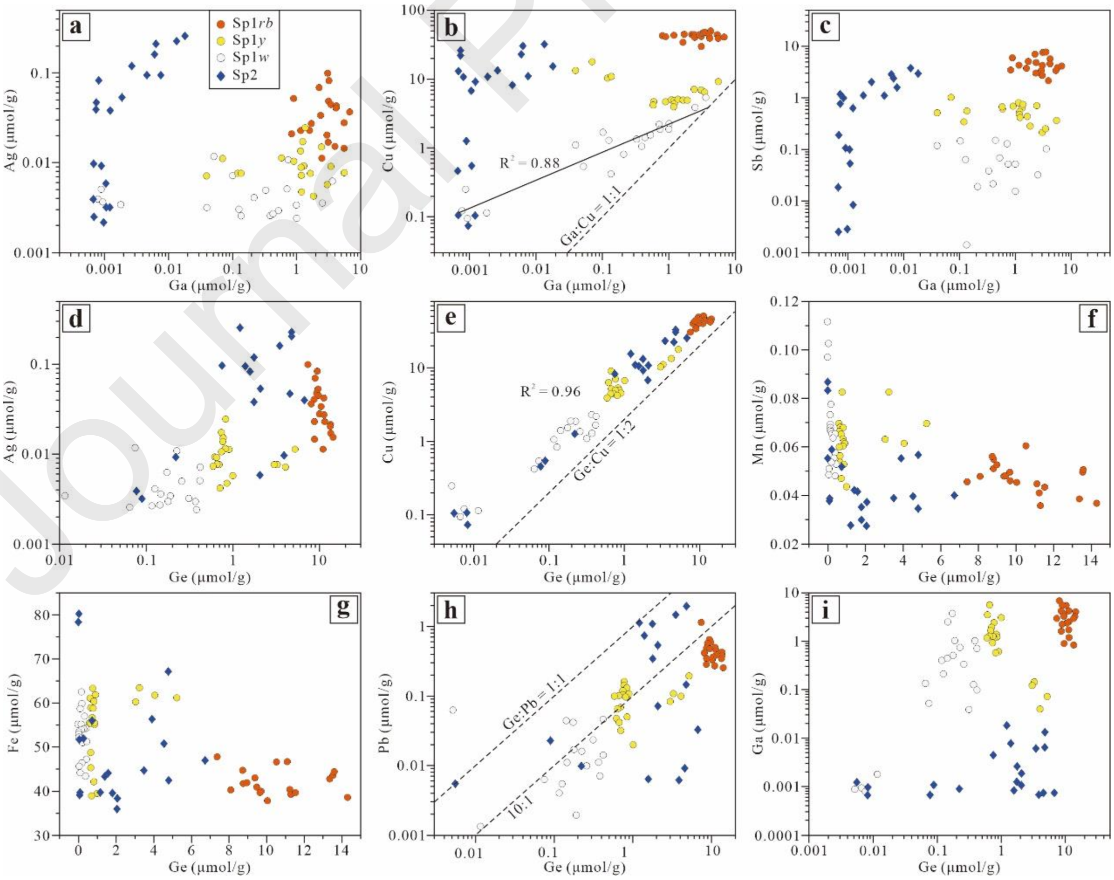

a
Sp1y
Sp1rb
Sp1w
250 µm
b
Fe
c
Cu
d
Ga
e
Ge
f
As
400
300
200
100
g
Cd
h
Sb
i
Pb
ppm

Fig. 7



Cal
Sp2
(101)
(001)
(011)
(010)
(110)
250 μm
ppm
Fe
Cu
Ga
Ge
As
Ag
Cd
In
Sb
Pb

Fig. 8



Sp1rb
Sp1y
Sp1w
Sp2
Ag (μmol/g)
Cu (μmol/g)
Sb (μmol/g)
Mn (μmol/g)
Fe (μmol/g)
Pb (μmol/g)
Ga (μmol/g)
Ge (μmol/g)
R² = 0.88
Ga:Cu = 1:1
R² = 0.96
Ge:Cu = 1:2
Ge:Pb = 1:1
10:1

Fig. 9

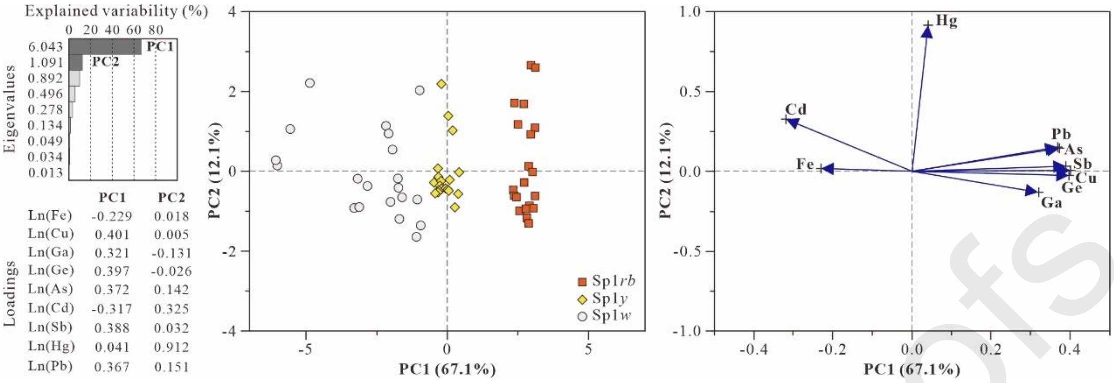


Fig. 10

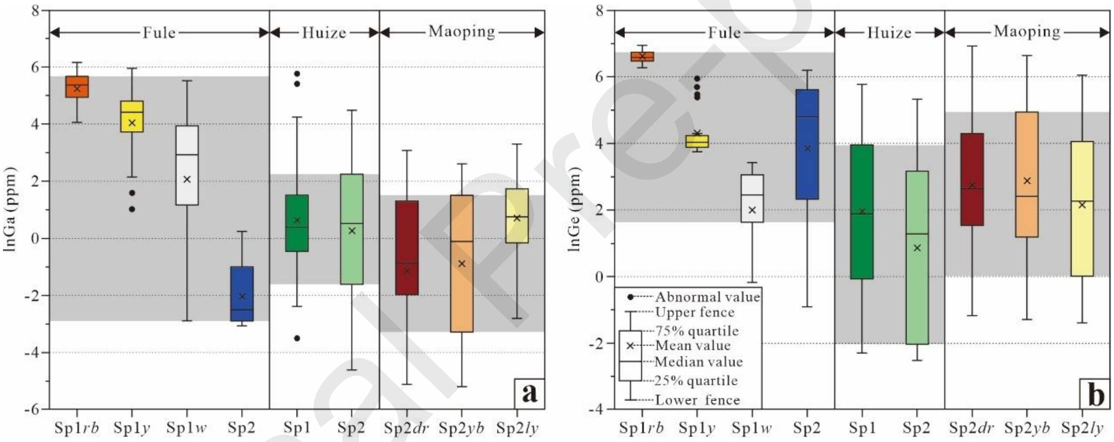


Fig. 11

Table 1

Sphalerite type	Mn	Fe	Cu	Ga	Ge	As	Ag	Cd	In	Sn	Sb	Hg	Tl	Pb
Sp1*rb* (*n*=20)														
Max.	3.31	2669	3143	474	1038	202	10.7	21476	1.38	16.5	929	224	0.11	239

Min.	1.98	2115	1919	58.2	535	77.6	1.22	13417	b.d.l.	b.d.l.	261	15.3	b.d.l.	53.5
Median	2.64	2330	2753	214	716	128	3.33	16346	0.10	0.83	516	24.2	0.02	87.0
Mean	2.60	2351	2709	218	760	131	4.04	16933	0.33	3.97	568	54.1	0.04	96.0
Q1	2.45	2221	2592	149	648	98.4	2.27	15184	0.01	0.12	418	17.7	0.00	72.6
Q3	2.79	2449	2858	280	824	154	4.89	18347	0.55	5.44	679	64.2	0.08	107
STD	0.34	156	285	108	140	37.3	2.48	2366	0.43	5.34	201	58.0	0.04	39.6
Sp1*y* (*n*=20)														
Max.	4.54	3531	1137	386	379	33.4	2.64	29044	0.44	5.86	123	117	0.08	41.0
Min.	2.40	2175	247	2.76	43.1	3.02	0.45	17248	b.d.l.	0.06	26.3	17.7	b.d.l.	4.18
Median	3.43	3221	314	83.5	56.1	10.2	0.98	21929		0.14	65.3	23.7		20.5
Mean	3.45	3082	435	101	100	11.8	1.09	22203		0.80	65.1	32.4		19.4
Q1	3.25	2995	285	41.1	49.5	6.35	0.81	20360		0.07	48.2	20.2		12.6

Q3	3.73	3422	487	118	66.7	14.6	1.23	24466		0.53	80.6	29.4		23.0
STD	0.52	432	230	91.1	95.8	7.68	0.50	2873		1.61	23.9	23.5		8.95
Sp1*w* (*n*=20)														
Max.	6.11	3487	340	251	30.6	4.20	1.27	40201	0.24	1.50	18.21	145	0.04	13.3
Min.	2.64	2426	5.98	b.d.l.	b.d.l.	b.d.l.	0.26	16111	b.d.l.	b.d.l.	b.d.l.	16.3	b.d.l.	0.07
Median	3.62	2987	82.9	19.0	11.7	1.53	0.39	23158		0.11	4.24	25.6	0.02	2.21
Mean	3.82	2948	92.4	41.5	13.2	1.71	0.50	26439		0.22	6.14	36.6	0.02	3.39
Q1	3.01	2800	32.5	3.42	5.25	1.01	0.32	19152		0.09	0.15	21.5	0.01	0.73
Q3	4.08	3107	118	50.2	19.8	1.97	0.55	33444		0.20	9.30	39.0	0.04	3.90
STD	1.00	288	80.4	62.3	9.90	0.84	0.27	8337		0.31	6.17	29.2	0.01	3.69
Sp2 (*n*=20)														
Max.	4.76	4483	2045	1.26	488	490	27.7	48492	b.d.l.	0.18	457	36.9	0.10	400

Min.	1.50	1995	4.65	b.d.l.	0.40	b.d.l.	0.23	9357	b.d.l.	b.d.l.	0.31	16.4	b.d.l.	0.06
Median	2.18	2478	681	0.08	122	124	4.66	21213			111	24.2	0.01	10.2
Mean	2.52	2752	743	0.25	150	145	7.92	23759			133	24.9	0.03	79.2
Q1	2.02	2210	68.4	0.06	13.6	15.8	0.58	17757			10.9	21.0	0.01	1.30
Q3	3.05	2961	1085	0.34	262	224	11.0	25660			206	25.7	0.04	122
STD	0.88	702	638	0.32	141	140	8.62	10001			137	5.48	0.03	118

Note: Max.—maximum; Min.—minimum; Mean—arithmetic mean; Q1—25% quartile; Q3—75% quartile; STD—standard deviation; b.d.l.—below detection limit.

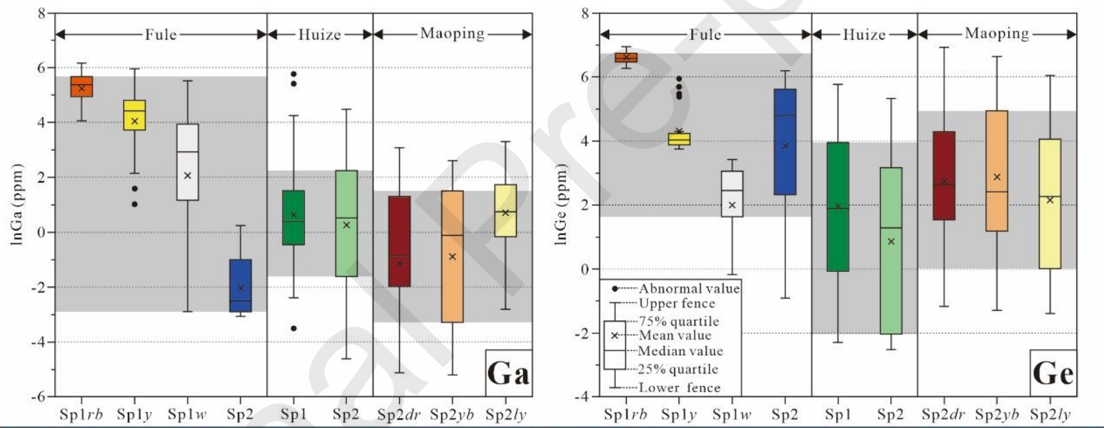
Table 2

	Fule deposit	Huize deposit	Maoping deposit
Scale and grade	10 Mt at 15–20 wt.% Pb + Zn	25 Mt at 30–35 wt.% Pb + Zn	19 Mt at 15–30 wt.% Pb + Zn
Associated dispersed elements	Cd, Se, Ga, and Ge	~1200 t Ge	~400 t Ge
Host strata	The middle Permian Yangxin Formation	The upper Ediacaran Dingying Formation and lower	The upper Devonian Zaige Formation and

		Carboniferous Baizuo Formation	upper Carboniferous Weining Formation
Host rock	Dolostone and limestone	Dolostone	Dolostone and limestone
Ore-controlling structure	NE-trending faults in the ore district	The NE-trending Kuangshanchang and Qilingchang fault	The NE-trending Maoping fault and SN-trending Luozehe fault
Spatial relationship with the Emeishan basalts	The orebody is spatially (< 1 m) close to the Emeishan basalts	The Emeishan basalts are distributed in the periphery of the ore district	The Emeishan basalts are distributed in the periphery of the ore district
Ore strcture	Disseminated, brecciated, massive, and veined	Massive, veined, disseminated, stockwork, and veined	Massive, brecciated, veined
Ore texture	Coarse- to fine-grained, replacement, fragmented, and solid solution	Coarse- to fine-grained, replacement, and fragmented, veinlet-vein filling	Replacement, co-dissolved, fragmented, euhedral-subhedral granular, and zoned
Mineralogy	Ore minerals: Sp, Gn, Py, Ccp, Ttr, Pld, Mil (Nic, Vae, Ger); Gangue minerals: Cal, Dol	Ore minerals: Sp, Gn, Py (Ccp); Gangue minerals: Cal, Dol, Brt	Ore minerals: Sp, Gn, Py (Ccp, Arg); Gangue minerals: Cal, Dol
Temperature of ore fluid	Sphalerite: mostly 220° to 235℃	Sphalerite: mostly 320° to 364℃ and 150° to 221℃	Sphalerite: mostly 240° to 300℃

Formation mechanism of reduced sulfur	Thermochemical sulfate reduction	Thermochemical sulfate reduction	Thermochemical sulfate reduction
References	Si et al. (2006), Lv et al. (2015), Zhou et al. (2018), Cui et al. (2018), Ren et al. (2019), Lyu et al. (2020), and Chen et al. (2022)	Han et al. (2014, 2016, 2023), Wang et al. (2016), Liu et al. (2022), Zhao et al. (2023), and Tan et al. (2023)	Ren et al. (2018), Wang et al. (2022, 2023a, b), and Han et al. (2023, 2024)

Abbreviations: Sp = sphalerite, Gn = galena, Py = pyrite, Ccp = chalcopyrite, Ttr = tetrahedrite, Pld = polydymite, Mil = millerite, Nic= niccolite, Vae = vaesite, Ger = gersdorffite, Arg = argentite, Cal = calcite, Dol = dolostone, Brt = barite.



Compared to sphalerite in other typical Pb-Zn deposits in the SYG region, sphalerite from the Fule deposit is more enriched in Ga and Ge. This suggests that the Emeishan basalts may have contributed additional Ga and Ge to this deposit.

- The highest Ga contents (mean 218 ppm, up to 474 ppm) and Ge contents (mean 760 ppm, up to 1038 ppm) are both found in reddish-brown sphalerite in stage I.

- The sphalerite color variations (reddish-brown, yellow, and white) at Fule are related to the contents of Cu, Ga, Ge, As, and Sb.

- The Emeishan basalts may have contributed additional Ga and Ge to the Fule deposit.